

Nicholas Lee

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Education

University of Waterloo – BAsC in Mechatronics Engineering, GPA: 3.85

Expected May 2028

Skills

Design & Manufacturing: SolidWorks, Onshape, nTop, parametric CAD, DFM/DFA, 3D printing, machining

Simulation & Analysis: COMSOL Multiphysics, ANSYS, CFD, FEA

Programming: Python, MATLAB/Simulink, Siemens TIA Portal, Git

Work Experience

Research Intern, Harvard University, Aizenberg Lab – Cambridge, MA

May 2026 – Present

- Demonstrated a potential geometry-driven selectivity mechanism for an intelligent electronic nose by simulating 10 chemical species in COMSOL and showing that each compound reaches peak concentration at a distinct channel location, enabling compound detection without modifying the sensor array
- Modeled the effects of bend angle and inlet velocity on sensor response through parametric CFD studies, identifying strong linear airflow trends and limited influence of bend geometry on detection
- Streamlined post-processing of transient simulations by automating extraction of flow and detection metrics and plot generation across all studies using Python (Pandas, NumPy, Matplotlib) to support engineering analysis
- Fabricated 3D-printed test geometries for benchtop validation of simulation results by converting CT-scanned animal nasal anatomies into printable STL models

Undergraduate Research Assistant, MSAM Lab – Waterloo, ON

Jan 2026 – Apr 2026

- Accelerated helical gear-design by developing a parametric SolidWorks model that supported rapid geometry changes across gear configurations, varying tooth count and pitch diameter
- Minimized material use in gearbox housing while preserving structural strength by designing internal lattice infill structures in nTop for additive manufacturing

Engineering Intern, Refresco Beverages – Mississauga, ON

Jan 2025 – Aug 2025

- Solved recurring contamination-related failures by redesigning a syrup-resistant pneumatic piston assembly in SolidWorks and integrating a replacement vendor piston
- Initiated a \$100K CAD CAPEX project to reduce worker injuries by scoping vacuum lift-assist solutions, evaluating vendor proposals, and presenting the business case to the plant manager
- Improved plant uptime by implementing preventive maintenance procedures and updating assembly drawings and bills of materials in the CMMS

Projects

Founding Mechanical Designer, Waterloo Automation Collective – Waterloo, ON

Jan 2026 – Present

- Led the clamping subsystem for a desktop injection molder from concept through CAD development to early prototyping, now in the machining stages
- Enabled rapid geometry iteration and centralized configuration management by developing parametrically driven clamping linkages in Onshape and SolidWorks

Mechanical Team Member, Midnight Sun Solar Car Team – Waterloo, ON

Sep 2025 – Present

- Reduced ballast box enclosure mass by 20% by designing a SolidWorks sheet-metal assembly, verifying structural load requirements through static FEA
- Designed surface-modeled cutting stencils that follow the aerobody's curvature, referenced to vehicle datums to standardize wheel-cover alignment and reduce installation errors
- Simplified rear-view camera installation and removal by designing and fit-testing a tool-free, press-fit 3D-printed mounting system